Software Requirements Specification (SRS)

Project: Smart College Club Management System

1. Introduction

Smart College Club Management System is a web-based platform that digitizes college club activities. It enables students to join clubs, register for events, coordinators to manage club operations, and administrators to oversee the complete system through role-based access.

2. Objectives

• Centralized club management
• Secure role-based authentication
• Event creation and registration
• QR-based attendance
• Automated email notifications
• Digital certificate management
• Reports and analytics

3. Users

Student
Club Coordinator
Administrator / Faculty

4. Functional Requirements

• User registration/login using Firebase Authentication
• Role-based dashboards
• Club creation and membership management
• Event creation, publishing and registration
• QR code attendance system
• Email notifications for confirmations and reminders
• Certificate generation and download
• Reports and analytics

5. Non-Functional Requirements

Security through Firebase Authentication, responsive UI, scalability, maintainability, performance, auditability and reliable database storage.

6. Technology Stack

Frontend: React.js
Backend: Spring Boot (Java)
Database: PostgreSQL/MySQL
Authentication: Firebase Authentication API
Email API: SendGrid
QR Library: ZXing

7. Database Tables

Users, Clubs, Club_Members, Events, Registrations, Attendance, Certificates, Notifications

8. External APIs

• Firebase Authentication API - Secure login and role-based access
• SendGrid Email API - Registration confirmation, reminders and certificate delivery

9. Future Enhancements

AI-based club recommendations, live chat, push notifications, geolocation-based attendance verification, mobile application, advanced analytics dashboard.